

Lecture Material #1

Information Security

Dr. Abu Nowshed Chy

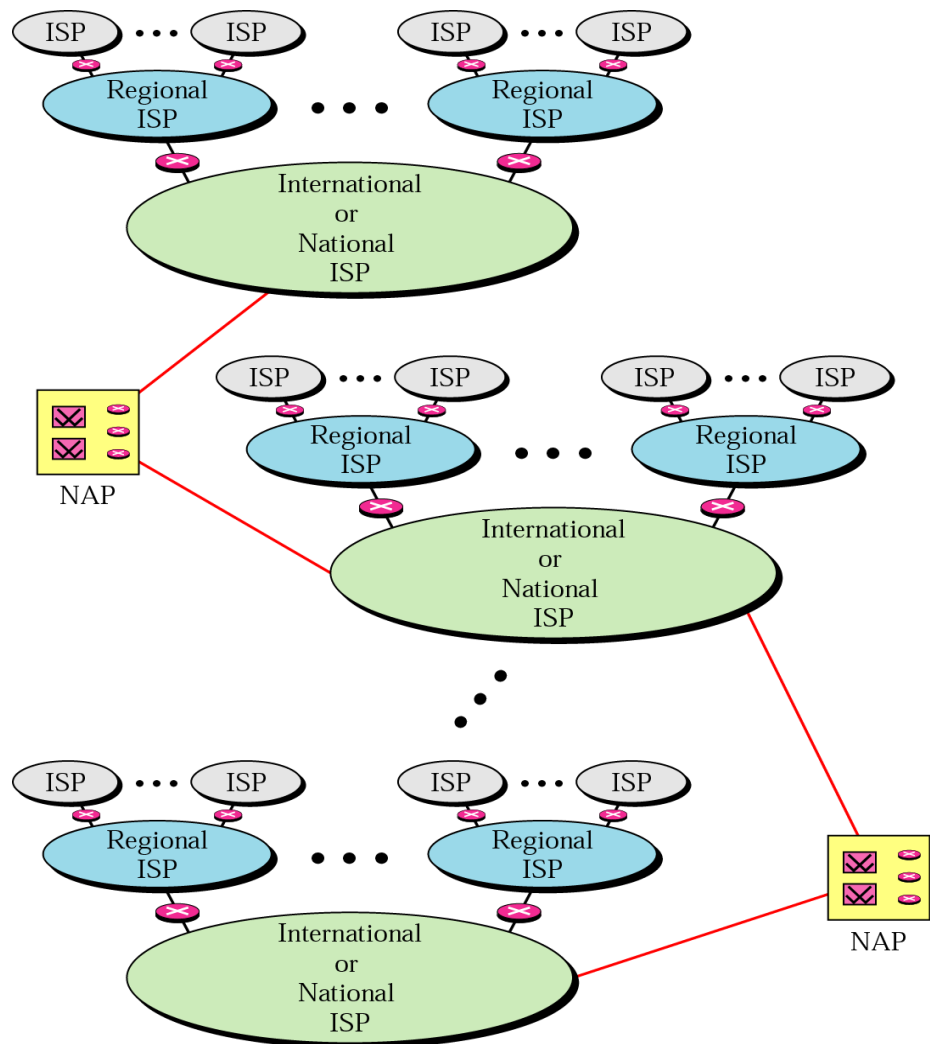
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WAN





Rogue Security Software





Rogue Security Software

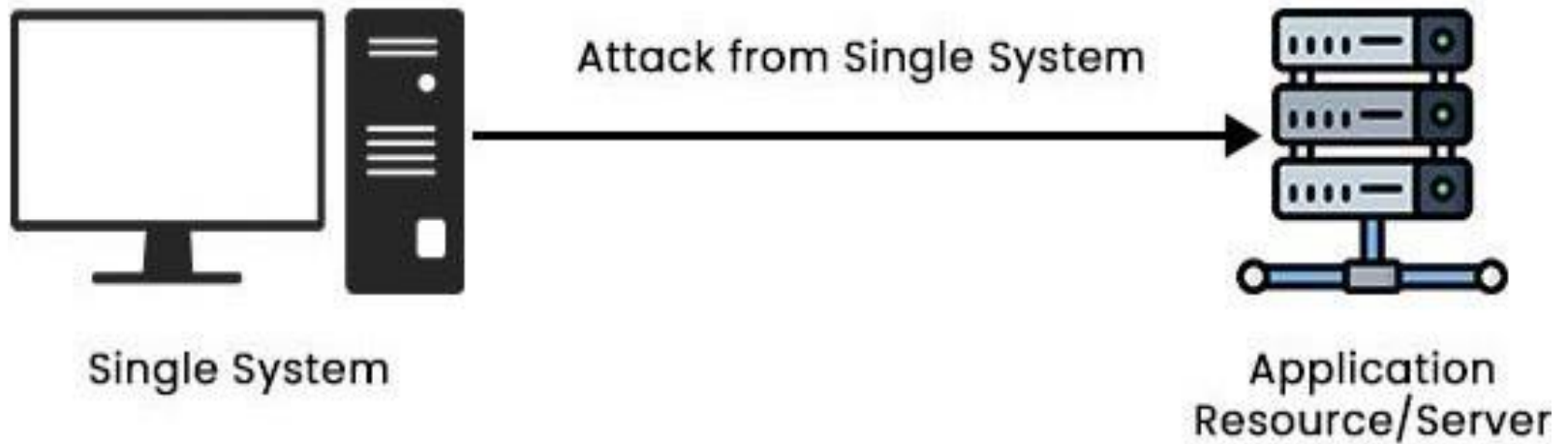
Leveraging the fear of computer viruses, scammers have found a new way to commit Internet fraud.

Rogue security software is **malicious software that mislead users to believe that they have network security issues**, most commonly a computer virus installed on their computer or that their security measures are not up to date. Then they offer to install or update users' security settings. They'll either **ask you to download their program to remove the alleged viruses**, or to pay for a tool. Both cases lead to actual malware being installed on your computer.





DOS Attack





DOS Attack

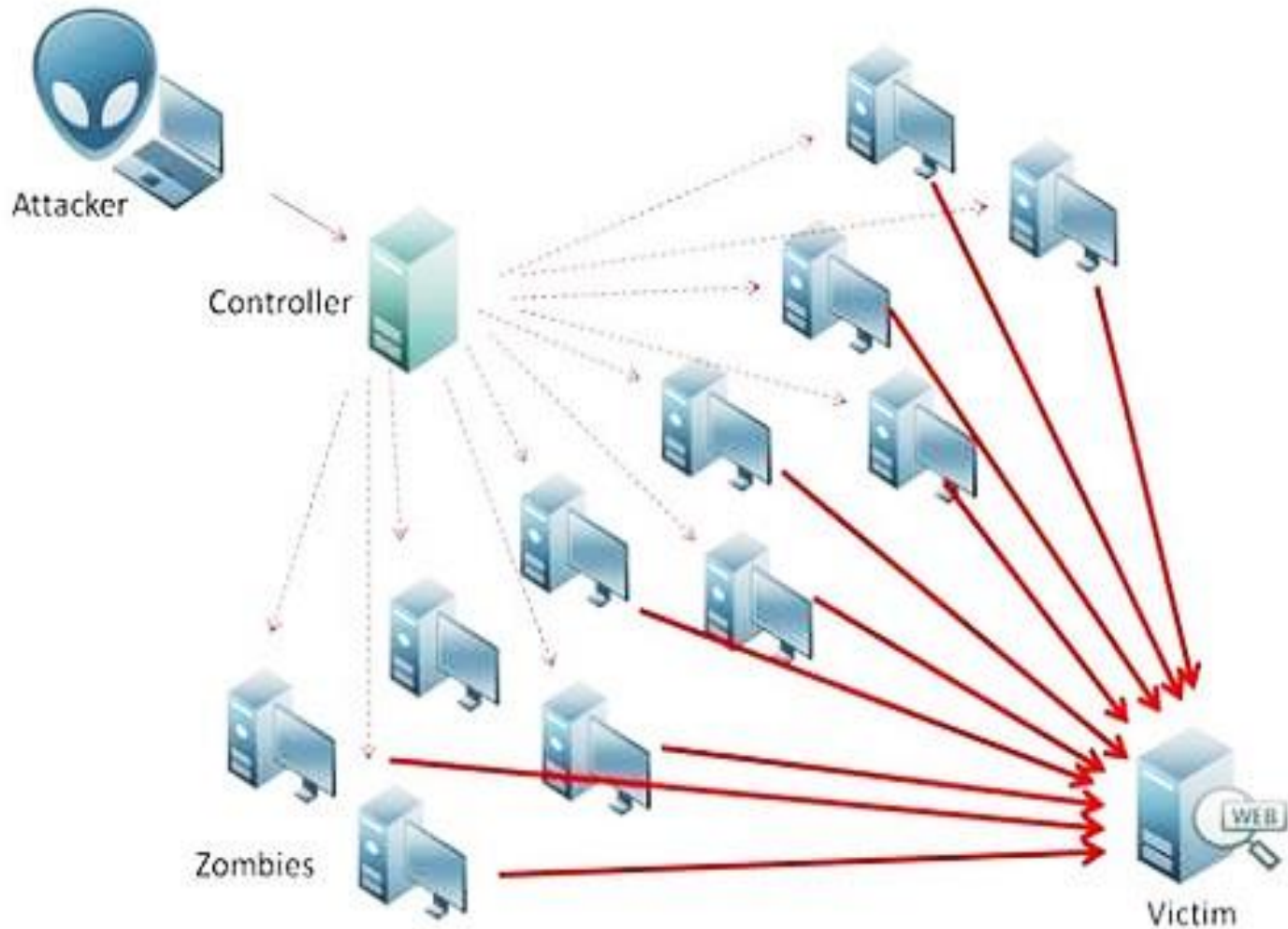
There are cases whenever a website's server gets overloaded with traffic and simply crashes, e.g. sometimes when a news story breaks. But more commonly, this is what happens to a website during a **DoS attack, or denial-of-service, a malicious traffic overload that occurs when attackers over-flood a website with traffic.** When a website has too much traffic, it's unable to serve its content to visitors.

A DoS attack is performed by one machine and its internet connection, by flooding a website with packets and making it impossible for legitimate users to access the content of flooded website. Fortunately, you can't really overload a server with a single other server or a PC anymore. In the past years it hasn't been that common if anything, then by flaws in the protocol.





DDOS Attack





DDoS Attack

A **DDoS attack**, or **distributed denial-of-service attack**, is similar to DoS, but is more forceful. It's harder to overcome a DDoS attack. It's **launched from several computers**, and **the number of computers involved can range from just a couple of them to thousands or even more**.

Since it's likely that not all of those machines belong to the attacker, they are compromised and added to the attacker's network by malware. These computers can be distributed around the entire globe, and that network of compromised computers is called botnet.

Since the attack comes from so many different IP addresses simultaneously, a **DDoS attack is much more difficult for the victim to locate and defend against**.



Phishing





Phishing

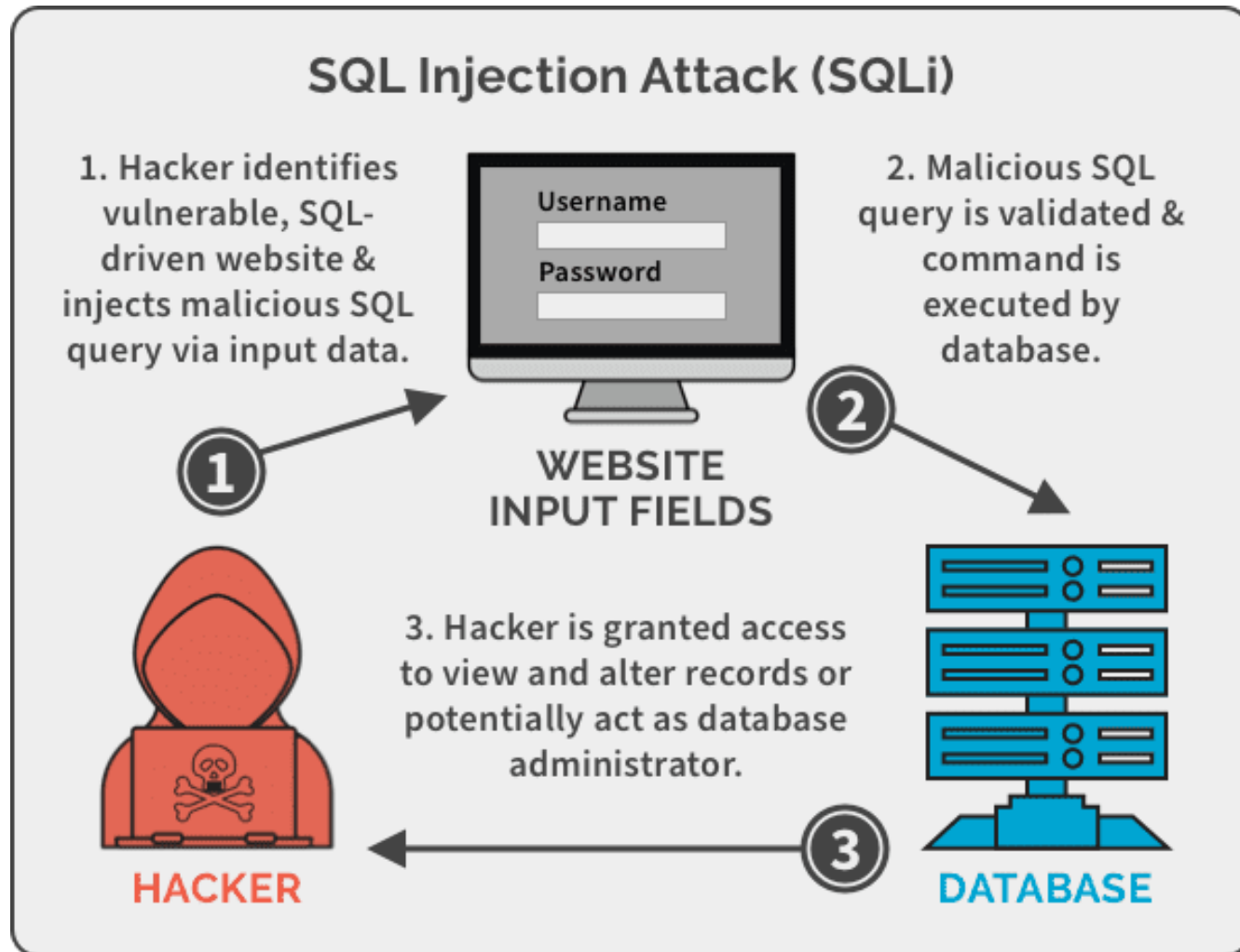
Phishing is a method of a social engineering with the goal of obtaining sensitive data such as passwords, usernames, and credit card numbers.

The attacks often come in the form of instant messages or phishing emails designed to appear legitimate. The recipient of the email is then tricked into opening a malicious link, which leads to the installation of malware on the recipient's computer. It can also obtain personal information by sending an email that appears to be sent from a bank, asking to verify your identity by giving away your private information.





SQL Injection Attack





SQL Injection Attack

```
txtUserId = getRequestString("UserId");  
txtSQL = "SELECT * FROM Users WHERE UserId = " + txtUserId;
```

UserId:

```
SELECT * FROM Users WHERE UserId = 105 OR 1=1;
```





SQL Injection Attack

`SELECT * FROM Users; DROP TABLE Suppliers`

```
txtNam = getRequestString("CustomerName");  
txtAdd = getRequestString("Address");  
txtCit = getRequestString("City");  
txtSQL = "INSERT INTO Customers (CustomerName,Address,City) Values(@0,@1,@2)";  
db.Execute(txtSQL,txtNam,txtAdd,txtCit);
```

```
$stmt = $dbh->prepare("INSERT INTO Customers (CustomerName,Address,City)  
VALUES (:nam, :add, :cit)");  
$stmt->bindParam(':nam', $txtNam);  
$stmt->bindParam(':add', $txtAdd);  
$stmt->bindParam(':cit', $txtCit);  
$stmt->execute();
```





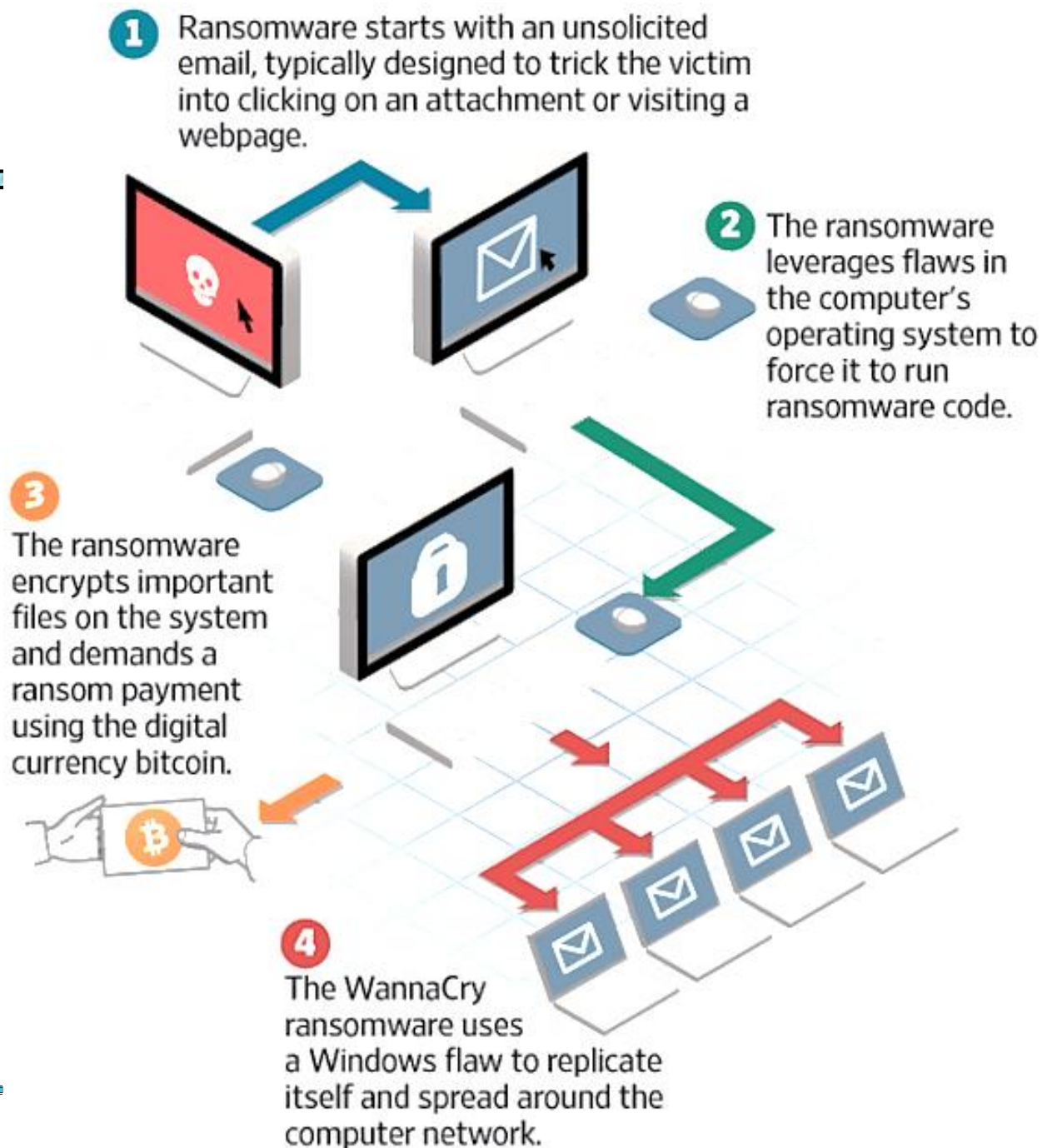
SQL Injection Attack

SQL injection attacks are designed to target data-driven applications by exploiting security vulnerabilities in the application's software. They use malicious code to obtain private data, change and even destroy that data, and can go as far as to void transactions on websites. It has quickly become one of the most dangerous privacy issues for data confidentiality.



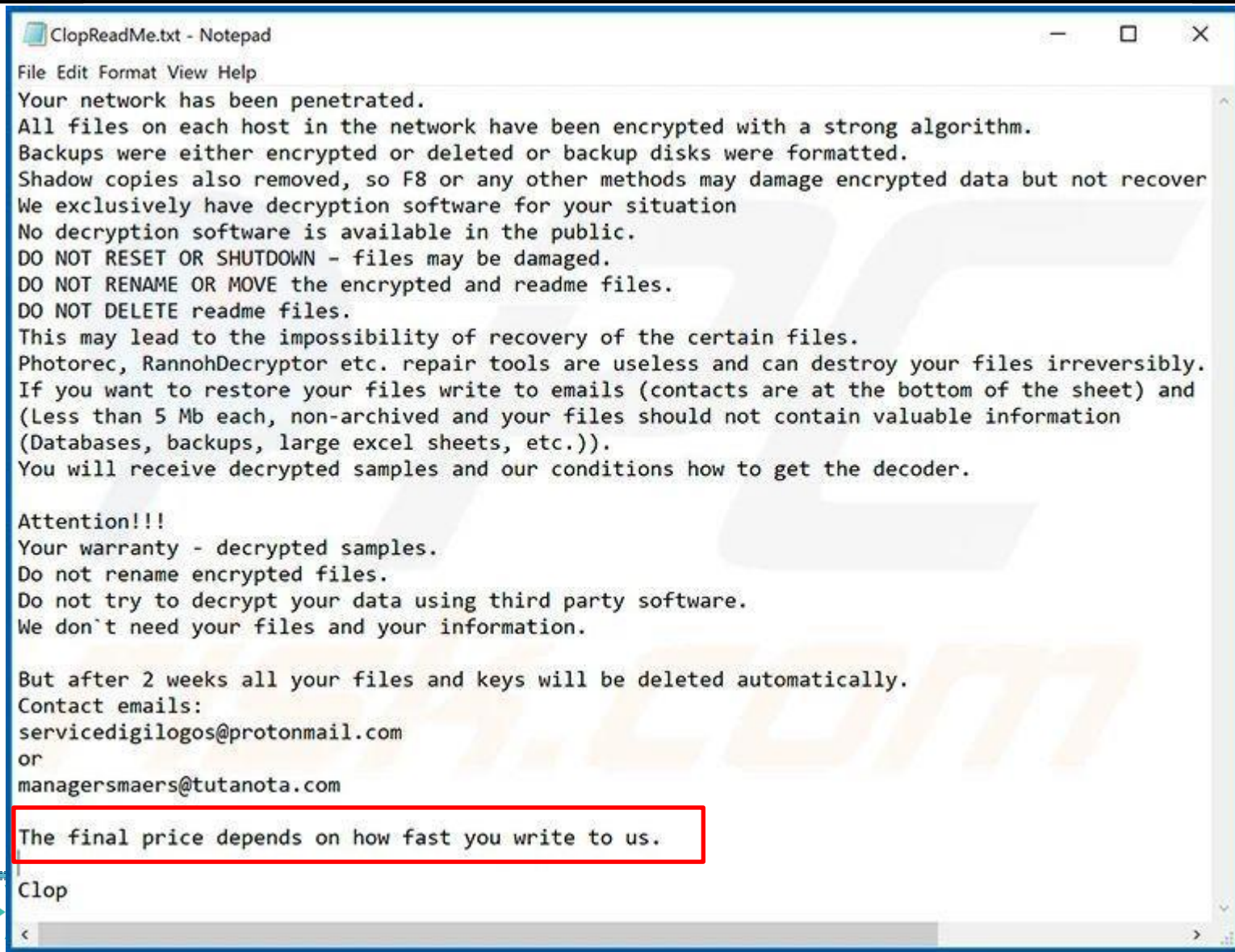


Ransomware





Ransomware





Ransomware

Ransomware is malware which encrypts your files until you pay a ransom to the hackers. “Clap” and “WannaCry” are the latest and most dangerous ransomware threats. It’s a variant of the well-known CryptoMix ransomware, which frequently targets Windows users.

Before beginning the encryption process, the Clap ransomware blocks over 600 Windows processes and disables multiple Windows applications, including Windows Defender and Microsoft Security Essentials — leaving you with zero chance of protecting your data.

The Clap ransomware has evolved since its inception, now targeting entire networks — not just individual devices. Even the 'Maastricht University' in the Netherlands became a victim of the Clap ransomware, with almost all Windows devices on the university’s network being encrypted and forced to pay a ransom.





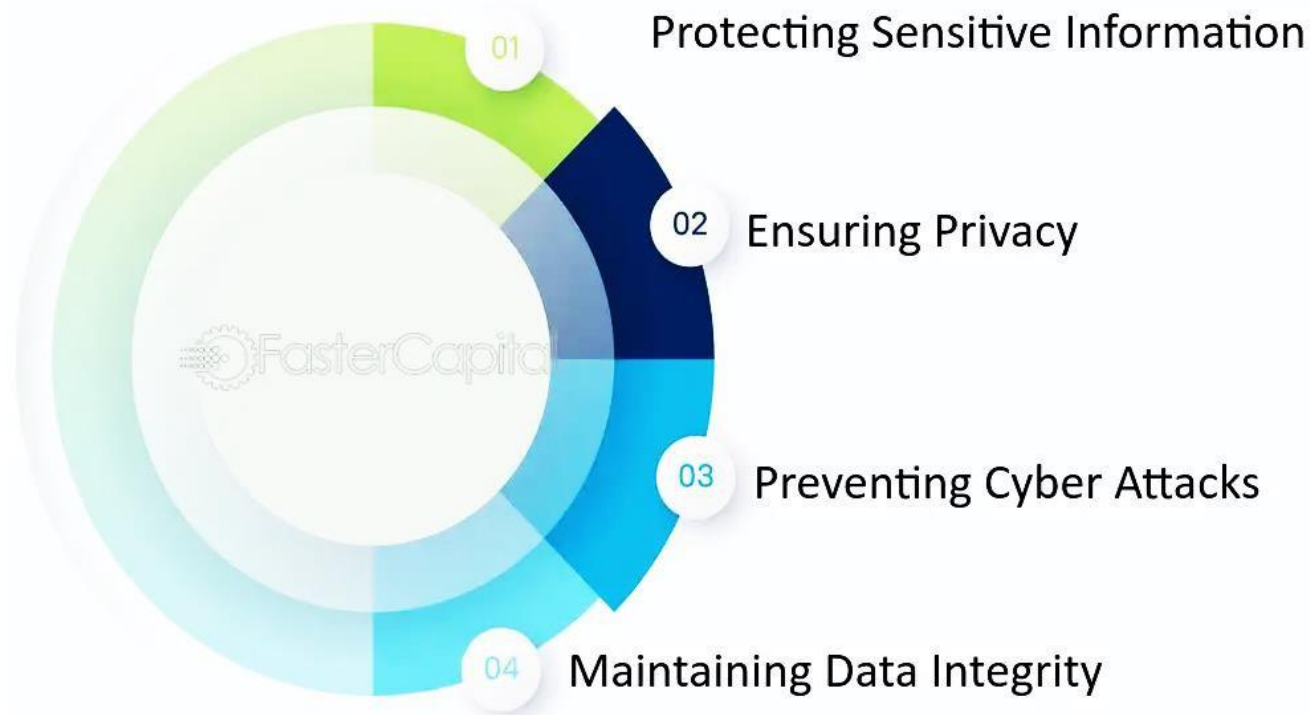
What are the Solutions!!!!





Importance of Cryptography

The Importance of Cryptography in Today's Digital World





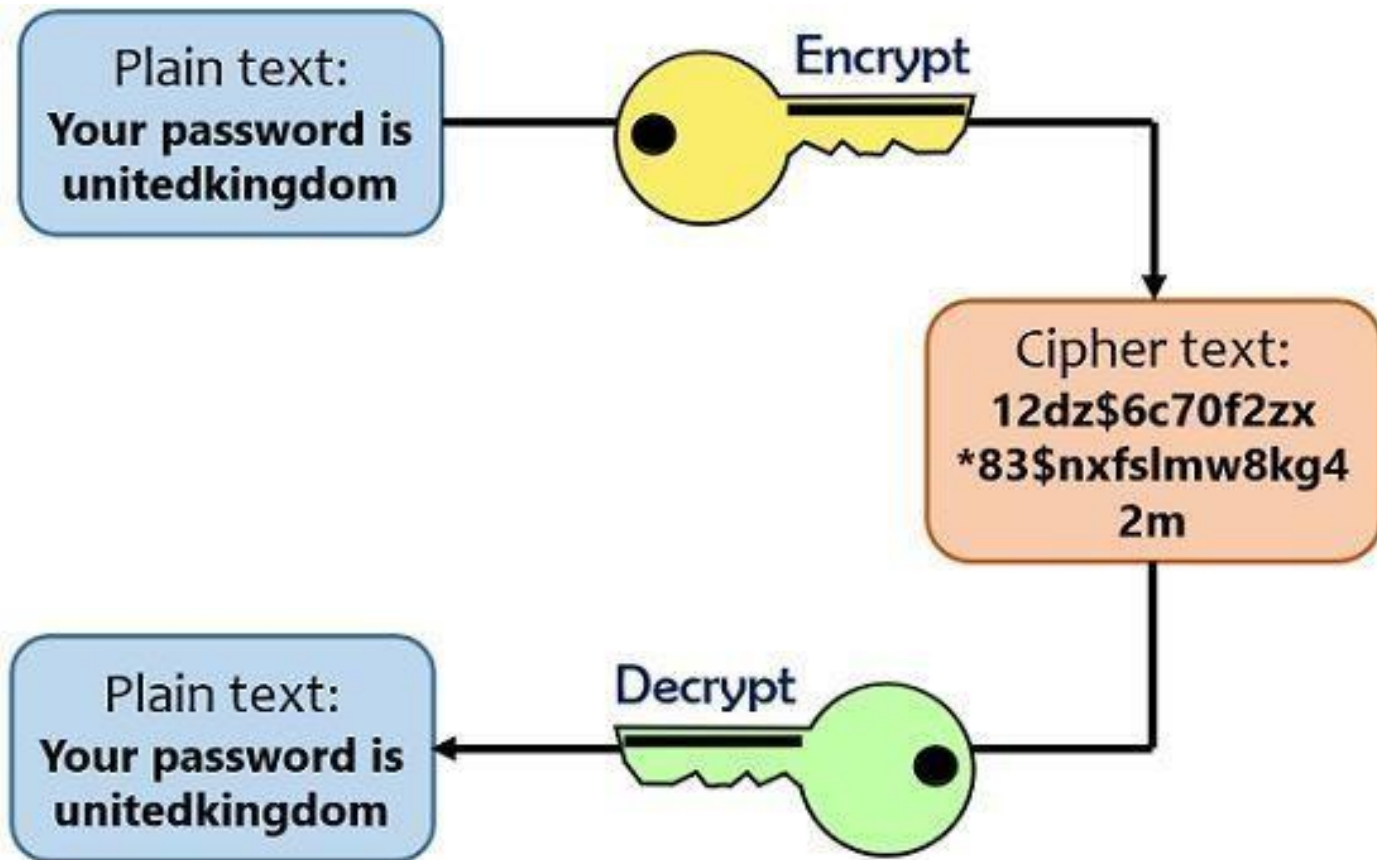
Cryptography



- ❖ Cryptography is the mathematical “**scrambling**” of data so that only someone with the necessary key can “**unscramble**” it.
- ❖ Cryptography allows **secure transmission of private information over insecure channels** (for example packet switched networks).
- ❖ Cryptography also allows **secure storage of sensitive data** on any computer.



Cryptography





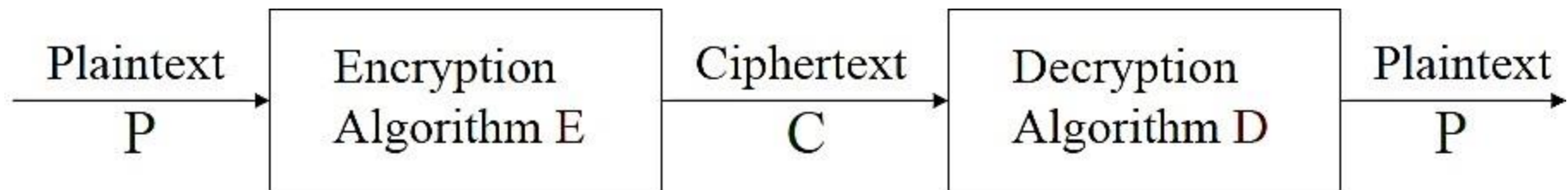
Common Terms in Cryptography

In cryptography, *encryption* is the process of transforming information (referred to as *plaintext*) using an algorithm (called *cipher*) to make it unreadable to anyone except those possessing special knowledge, usually referred to as a *key*. The result of the process is encrypted information (in cryptography, referred to as *ciphertext*). In many contexts, the word encryption also implicitly refers to the reverse process, *decryption*, to make the encrypted information readable again (i.e. to make it *unencrypted*).





Cryptosystem



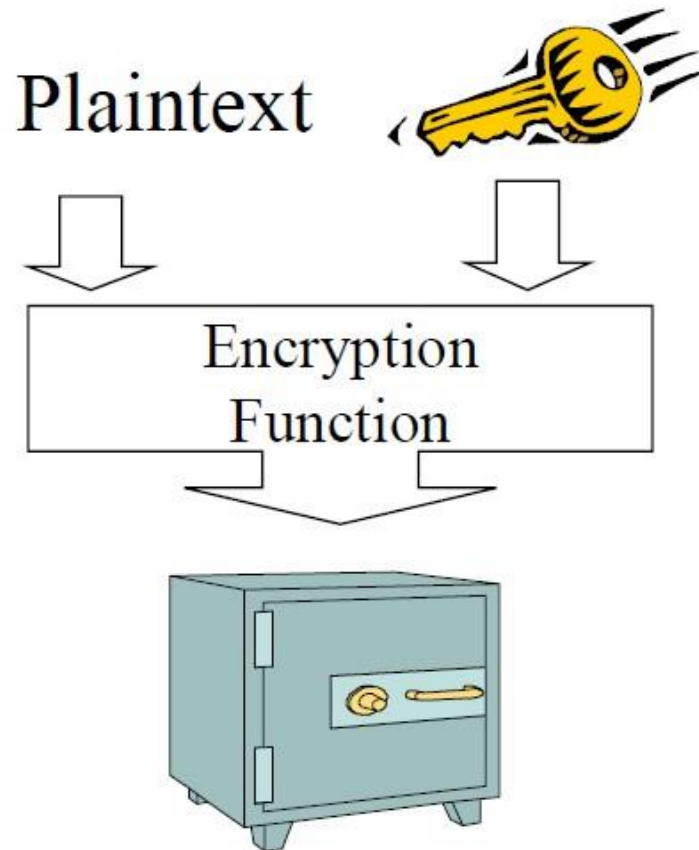
- Algorithm: a set of rules
- Algorithm often uses a parameter – key K
 - Key provides flexibility
 - Same algorithm can be used by different users with different keys
 - Security depends on the secret key value while algorithm can be public





Encryption

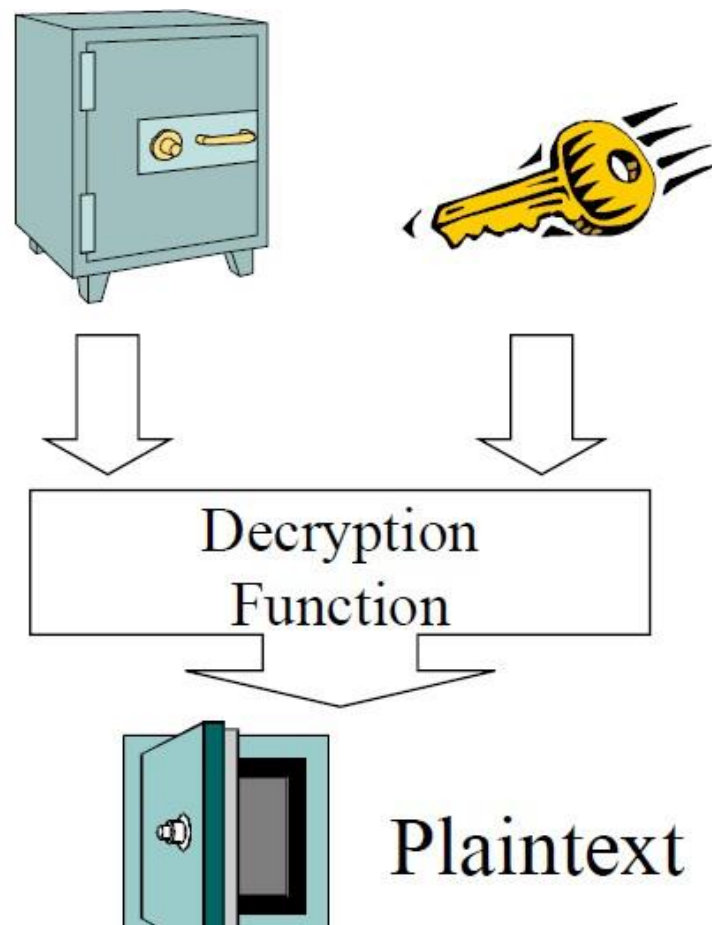
- Encryption is the process of feeding plaintext and key into a function and getting ciphertext output
- Ciphertext is “garbage” unless decrypted



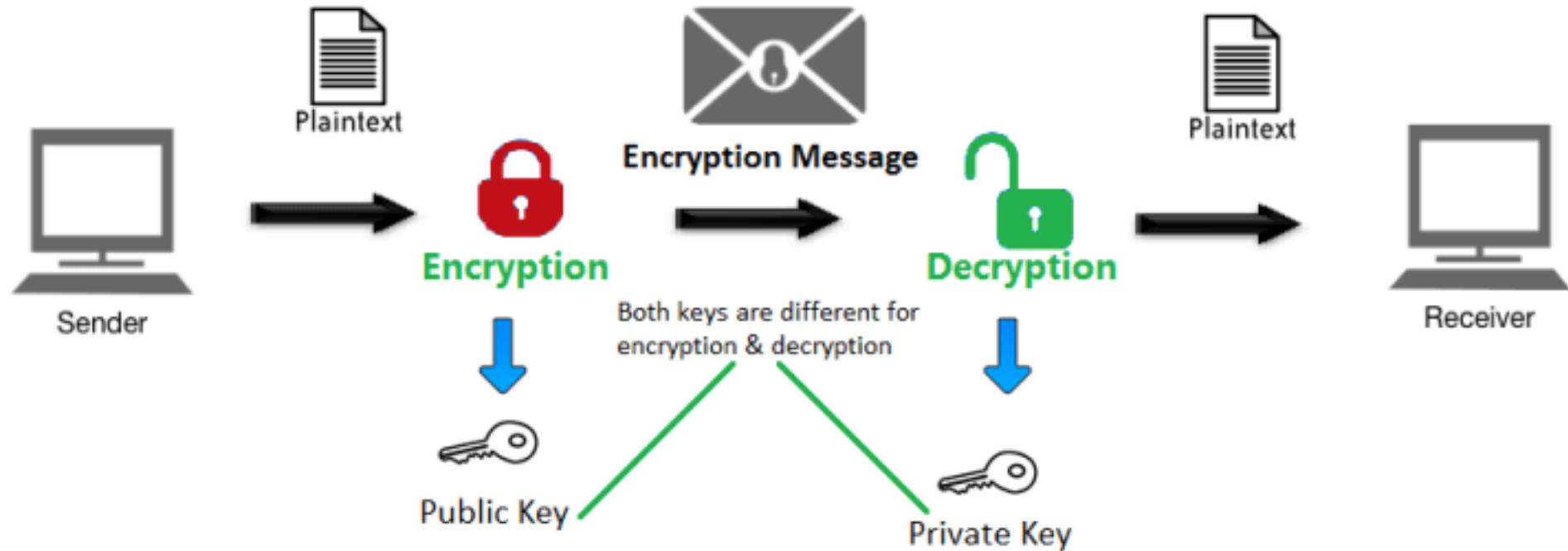


Decryption

- Decryption is the process of feeding ciphertext and a key into another function and getting original plaintext output



How Cryptography Works with Public and Private Key?





How Cryptography Works with Public and Private Key?

Imagine Alice wants to receive secret messages.

Step 1: Keys

- Alice has:
 -  Public key → she shares this with everyone
 -  Private key → she keeps this secret

Think of:

- Public key = an open padlock
- Private key = the only key that can open that padlock



How Cryptography Works with Public and Private Key?



Step 2: Sending a message

- Bob wants to send a secret message to Alice.
- Bob takes Alice's **public key (padlock)** and locks his message.
- Once locked, **nobody (not even Bob)** can open it.

Step 3: Receiving the message

- Alice uses her **private key** to unlock and read the message.

 Even if someone intercepts the message, they **cannot read it** without Alice's private key.





Common Cryptography Techniques!!!!





Modular Arithmetic



If a and b are integers with $a \neq 0$, we say that a *divides* b if there is an integer c such that $b = ac$. When a divides b we say that a is a *factor* of b and that b is a *multiple* of a . The notation $a \mid b$ denotes that a divides b . We write $a \nmid b$ when a does not divide b .

Determine whether $3 \mid 7$ and whether $3 \mid 12$.





Modular Arithmetic

Let a , b , and c be integers. Then

- (i) if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
- (ii) if $a \mid b$, then $a \mid bc$ for all integers c ;
- (iii) if $a \mid b$ and $b \mid c$, then $a \mid c$.





Modular Arithmetic

THE DIVISION ALGORITHM Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$.

In the equality given in the division algorithm, d is called the *divisor*, a is called the *dividend*, q is called the *quotient*, and r is called the *remainder*. This notation is used to express the quotient and remainder:

$$q = a \text{ div } d, \quad r = a \text{ mod } d.$$

What are the quotient and remainder when 101 is divided by 11?

What are the quotient and remainder when -11 is divided by 3?





Caesar Cipher Encryption

To express Caesar's encryption process mathematically, first replace each letter by an integer from 0 to 25, based on its position in the alphabet. For example, replace *A* by 0, *K* by 10, and *Z* by 25. Caesar's encryption method can be represented by the function f that assigns to the nonnegative integer p , $p \leq 25$, the integer $f(p)$ in the set $\{0, 1, 2, \dots, 25\}$ with

$$f(p) = (p + 3) \bmod 26.$$

A = 0	B = 1	C = 2	D = 3	E = 4	F = 5	G = 6	H = 7	I = 8	J = 9
K = 10	L = 11	M = 12	N = 13	O = 14	P = 15	Q = 16	R = 17	S = 18	T = 19
U = 20	V = 21	W = 22	X = 23	Y = 24	Z = 25				

What letter replaces the letter word “CSECU” when the function $f(p) = (7p + 3) \bmod 26$ is used for encryption?





Caesar Cipher Encryption

A = 0	B = 1	C = 2	D = 3	E = 4	F = 5	G = 6	H = 7	I = 8	J = 9
K = 10	L = 11	M = 12	N = 13	O = 14	P = 15	Q = 16	R = 17	S = 18	T = 19
U = 20	V = 21	W = 22	X = 23	Y = 24	Z = 25				

What letter replaces the letter K when the function
 $f(p) = ((7p*7) + 3) \bmod 26$ is used for encryption?





Caesar Cipher Decryption

$$f^{-1}(p) = (p - k) \bmod 26.$$

Decrypt the ciphertext message “PBZOBQ” that was encrypted using the Caesar cipher encryption technique with shift $k = -3$.

A = 0	B = 1	C = 2	D = 3	E = 4	F = 5	G = 6	H = 7	I = 8	J = 9
K = 10	L = 11	M = 12	N = 13	O = 14	P = 15	Q = 16	R = 17	S = 18	T = 19
U = 20	V = 21	W = 22	X = 23	Y = 24	Z = 25				





Caesar Cipher Decryption

What is the secret message produced from the message “CSECU” using the Caesar cipher with the encryption function $f(p) = (p + 12) \bmod 26$?

Decrypt the ciphertext message “CHATTOGRAM” that was encrypted using the Caesar cipher encryption technique with shift $k = 7$.

A = 0	B = 1	C = 2	D = 3	E = 4	F = 5	G = 6	H = 7	I = 8	J = 9
K = 10	L = 11	M = 12	N = 13	O = 14	P = 15	Q = 16	R = 17	S = 18	T = 19
U = 20	V = 21	W = 22	X = 23	Y = 24	Z = 25				





Hill Cipher





Hill Cipher Encryption

A = 0	B = 1	C = 2	D = 3	E = 4	F = 5	G = 6	H = 7	I = 8	J = 9
K = 10	L = 11	M = 12	N = 13	O = 14	P = 15	Q = 16	R = 17	S = 18	T = 19
U = 20	V = 21	W = 22	X = 23	Y = 24	Z = 25				

$$C = (K \times P) \bmod 26$$

Where:

- P = plaintext vector (from alphabets)
- K = key matrix
- C = ciphertext vector





Hill Cipher Encryption

What letter replaces the word “HI” when using the Hill Cipher encryption with the following key matrix?

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$





Hill Cipher Encryption

What letter replaces the word “HI” when using the Hill Cipher encryption with the following key matrix?

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$H = 7$$

$$I = 8$$

$$C = (K \times P) \bmod 26$$

$$C = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$K \times P = \begin{bmatrix} 3 \times 7 + 3 \times 8 \\ 2 \times 7 + 5 \times 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$





Hill Cipher Encryption

What letter replaces the word “HI” when using the Hill Cipher encryption with the following key matrix?

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$H = 7$$

$$I = 8$$

$$P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} 45 \\ 54 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 2 \end{bmatrix}$$

$$19 = \mathbf{T}$$

$$2 = \mathbf{C}$$





Hill Cipher Encryption

Example: Plain text is CD. Find out cipher text using hill cipher. Encrypt Plain text using hill cipher.

$$\text{Key Matrix} = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$$

Solution:

❑ Plain Text = CD (C = 2, D = 3)

$$\text{Key matrix} = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$$

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

$$\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} * \begin{bmatrix} C \\ D \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} * \begin{bmatrix} 2 \\ 3 \end{bmatrix} \rightarrow \begin{bmatrix} 13 \\ 18 \end{bmatrix} \% 26 \rightarrow \begin{bmatrix} 13 \\ 18 \end{bmatrix} \rightarrow \begin{bmatrix} N \\ S \end{bmatrix}$$

$$\rightarrow ((2 * 2) + (3 * 3)) \rightarrow 4 + 9 \rightarrow 13$$

$$\rightarrow ((3 * 2) + (4 * 3)) \rightarrow 6 + 12 \rightarrow 18$$

Cipher Text = NS





Playfair Cipher





Playfair Cipher

Step 1: Choose Key phrase for the Encryption of your Plain text.

Let's say we want to encrypt the plain text "HELLO WORLD" using the key phrase "**SECURITY**".

When choosing a term, make sure that **no letter is duplicated**, and especially that the **letters I and J do not appear together**. Keywords like INJURE, JUICE, and JIGSAW, for example, would be disqualified since they feature both I and J at the same time, which violates this criteria.





Playfair Cipher

Step 2: Now create **5 X 5 Playfair Matrix** or Key Matrix according to your Key.

S	E	C	U	R
I	T	Y		





Playfair Cipher

The remaining squares of the matrix are then filled with the remaining alphabet letters, in alphabetical sequence. However, since there are 26 letters and only 25 squares available, we allocate both I and J to the same square.

S	E	C	U	R
I/J	T	Y	A	B
D	F	G	H	K
L	M	N	O	P
Q	V	W	X	Z





Playfair Cipher



Step 3: Preparing the Message

The next step is to prepare the message for encryption. We **remove any non-alphabetic characters and convert all letters to uppercase**. In our example, the message “HELLO WORLD” becomes “**HELLOWORLD**”.





Playfair Cipher

Step 4: Breaking the Message into Pairs. (Making digraphs of Plaintext)

To encrypt the message using the Playfair cipher, we **break the message into pairs of letters**. If we have an **odd number of letters**, we usually **add a placeholder letter (e.g., "X")** at the end to create a pair. If **in between duplicate letters** come in same pair then we will break it into single letter and **attach X** at the first letter then will start making the other pairs.

"HELLOWORLD"

"HE | LX | LO | WO | RL | DX".

"FU | OQ | MP | XN | SP | HQ".





Playfair Cipher

Step 5: Now take one pair at a time and look for it in the key matrix.

There are mainly **three criteria** for encrypting letters.

#1: If the two letters in the pair are in the same row of the key square, we replace them with the letter to their right.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z





Playfair Cipher

#2: If both letters in the pair are found in the same column of the key square, we will replace each letter with the letter below it.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z





Playfair Cipher

#2: If both letters in the pair are found in the same column of the key square, we will replace each letter with the letter below it.

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

DE

Shape: Column

Rule: Pick Items Below Each
Letter, Wrap to Top if Needed

OD





Playfair Cipher

#3: If the letters are in different rows and columns, we form a rectangle with them and change each letter with the letter in the opposite corner. Let suppose we have to find encrypted text for N and T.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z





Playfair Cipher

#3: If the letters are in different rows and columns, we form a rectangle with them and change each letter with the letter in the opposite corner. Let suppose we have to find encrypted text for N and T.

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z





Playfair Cipher

S	E	C	U	R
I/J	T	Y	A	B
D	F	G	H	K
L	M	N	O	P
Q	V	W	X	Z

“HELLOWORLD”

“HE | LX | LO | WO | RL | DX”.

“FU | OQ | MP | XN | SP | HQ”.





Playfair Cipher (Practice Example)



Let's say we want to **encrypt** the plain text "HELLO WORLD" using the key phrase "KEYWORD."





Playfair Cipher (Practice Example)

K	E	Y	W	O
R	D	A	B	C
F	G	H	I/J	L
M	N	P	Q	S
T	U	V	X	Z

“HELLOWORLD”

“HE | LX | LO | WO | RL | DX”

“GY | JZ | SC | OK | CF | BU”





Playfair Cipher (Practice Example)

Using the following Playfair matrix, encrypt the following message:

See you at CSECU

M	F	H	I/J	K
U	N	O	P	Q
Z	V	W	X	Y
E	L	A	R	G
D	S	T	B	C





Playfair Cipher (Practice Example)

Let's say we want to **decrypt** the plain text
“gddogdrqprsdhmembv” using the key phrase
“Charles”.

C	H	A	R	L
E	S	B	D	F
G	I/J	K	M	N
O	P	Q	T	U
V	W	X	Y	Z



